

# Multi-Functional Interface Passivation via Guanidinium Iodide Enables Efficient Perovskite Solar Cells

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Perovskite solar cells have become a leading contender in next-generation photovoltaic technologies due to their high efficiency and low-cost potential. Managing the deep defects present effectively in the crystal lattice and at the interfaces is essential for enhancing the performance and longevity of perovskite solar cells. Here, perovskite's crystallization modulation and interfacial defect passivation are achieved by developing a guanidinium iodide (GAI)-based surface passivation strategy. The integration of GAI passivates the grain boundaries, leading to a perovskite thin film with a smoother and more uniform grain distribution, facilitating charge carrier transport. Notably, the ammonium group, unsaturated nitrogen atoms, and iodide ions in GAI can collectively repair the surface defects of perovskite through various pathways, effectively suppressing non-radiative recombination, thereby enhancing the photovoltaic performance of the device. Ultimately, the champion device treated with GAI achieves a power conversion efficiency (PCE) of 23.02% and demonstrates similar ambient stability under unencapsulated conditions. These findings underscore the effectiveness of GAI passivation as a strategy to balance the improvement of the performance and stability of perovskite solar cells.

been widely recognized, particularly for the significant progress in enhancing solar cells' power conversion efficiency (PCE). These materials, especially the organic-inorganic hybrid metal-halide perovskites, exhibit unique optoelectronic properties, such as low exciton binding energy,<sup>[2]</sup> high carrier mobility,<sup>[3]</sup> long carrier diffusion lengths,<sup>[4]</sup> strong light absorption, and a tunable bandgap. These properties have dramatically increased the PCE of single-junction PSCs from 3.8% to over 26% in just a decade, with some silicon tandem devices exceeding 33%.<sup>[5]</sup> Despite these advancements, the long-term stability of PSCs remains a significant challenge, a major hindrance to their commercialization.

High defect densities at grain boundaries and interfaces in PSCs significantly decline performance and reduce longevity.<sup>[6,7]</sup> Solution-based synthesis of perovskite films introduces defects such as halide and cation vacancies and Pb-I antisite defects, which act as non-radiative recombination centers

and lower photovoltaic efficiency.<sup>[8–10]</sup> To address this issue, extensive research has focused on producing PSCs with minimal defect density, where interface passivation has proven to be a key strategy in defect reduction.<sup>[11–13]</sup> Guanidinium (GA)-based materials, particularly guanidinium iodide (GAI), exhibit the capability to passivate defects due to their electron-donating amino groups and hydrogen bonding abilities, coordinating iodide ions and increasing defect formation energy, thus enhancing perovskite stability.<sup>[14–16]</sup> Studies have shown that GA-based additives can effectively suppress iodide vacancies and non-radiative recombination.<sup>[17]</sup> Furthermore, guanidinium ion (GA<sup>+</sup>) doping has been found to improve photovoltaic performance and modify hysteresis in solar cells. While low concentrations of GA<sup>+</sup> can lead to lattice expansion and improved energy level alignment, higher concentrations may result in phase segregation and interface recombination.<sup>[18]</sup> Despite the success of GAI as an additive and buried interface layer in passivating perovskite defects, its hydrophilicity poses a challenge to the ambient stability of the devices, limiting the exploration of GAI as a top interface modification layer. Therefore, investigating the impact of GAI as a top interface passivation layer on device efficiency and stability while considering its hydrophilic properties is essential to comprehending GAI's passivation mechanism thoroughly.

## 1. Introduction

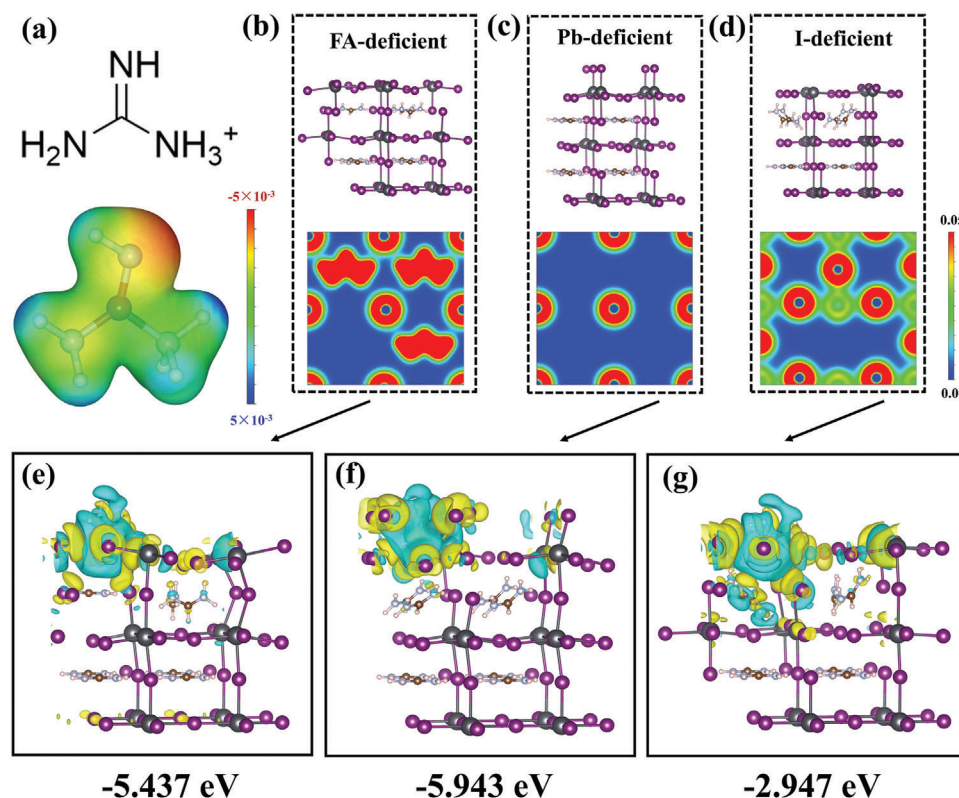
Perovskite solar cells (PSCs), an emerging photovoltaic technology, are rapidly gaining attention in renewable energy due to their cost-effectiveness, lightweight nature, and advantages in manufacturing flexibility. Since the initial report on perovskite materials in 2009,<sup>[1]</sup> their potential for photovoltaic applications has

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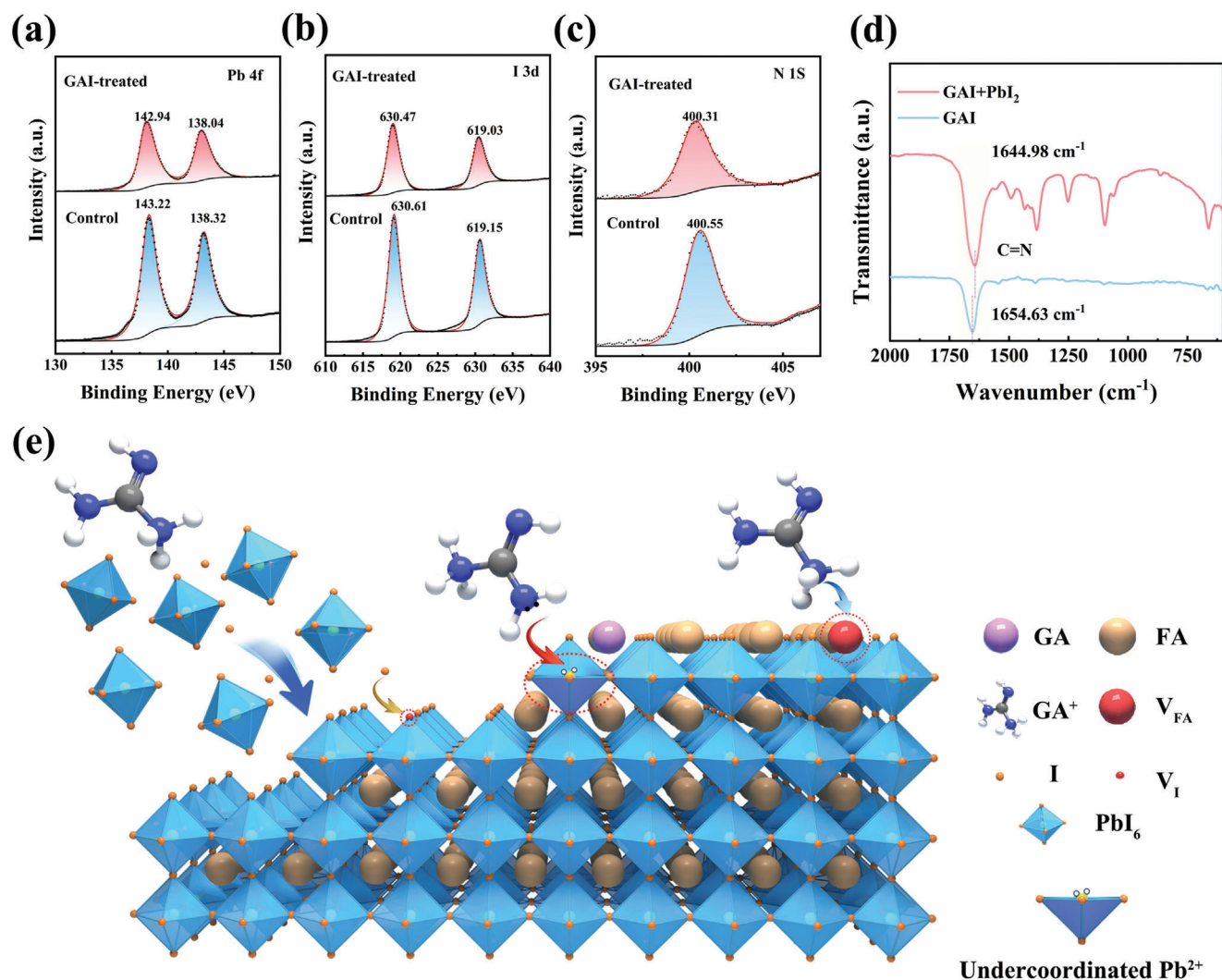
**Figure 1.** Electrostatic potential maps and binding energies. a) Chemical structure (top) and electrostatic potential map (bottom) of the GA<sup>+</sup> molecule. b) FA-deficient, c) Pb-deficient surface, and d) I-deficient surface studied by DFT and corresponding top views of the obtained electron density profiles (scale bar on the right). Electron density differences for the GA<sup>+</sup> molecule on the e) FA-deficient, f) Pb-deficient, and g) I-deficient surfaces, and resultant binding energies (Pink, brown, light gray, violet, and gray balls represent the H, C, N, I, and Pb atoms, respectively).

Herein, we aim to design a global passivation strategy to modulate the crystallization process and passivate interfacial defects in perovskite solar cells. Therefore, we propose multi-functional passivation molecules of GAI, consisting of electron-rich and electron-poor domains. We find that the GAI effectively modulates the crystallization process and passivates interfacial defects in perovskite solar cells, with the perovskite composition being formamidinium lead triiodide (FAPbI<sub>3</sub>) with an additional 0.8 mol% incorporation of methylammonium lead tribromide (MAPbBr<sub>3</sub>). An optimal amount of GAI effectively passivates grain boundaries, resulting in smoother and more uniform perovskite thin films with a refined grain distribution. To gain fundamental insights into the passivation mechanism of GAI, we theoretically constructed three different perovskite surfaces, including an A-site cation vacancy defect surface, a lead-deficient surface, and an iodine-deficient surface. Density functional theory (DFT) calculations show that GAI molecules can effectively passivate all these possible surface dangling bonds, independent of the chemical and electronic properties of the surface defects. We demonstrate the multi-functional passivation mechanisms of the ammonium groups, unsaturated nitrogen atoms, and iodide ions. This synergistic repair of surface defects effectively suppresses non-radiative recombination, enhancing the photovoltaic performance. Devices treated with the optimal concentration of 3 mg mL<sup>-1</sup> GAI achieved the highest efficiency of 23.02%, whereas higher concentrations of GAI led to a decline in device

performance. Moreover, despite the hydrophilic nature of GAI, devices treated with optimized GAI exhibited no observable difference in water stability under ambient conditions with a relative humidity of 30–40%.

## 2. Results and Discussion

We conducted DFT calculations to understand the interaction between GAI and different perovskite surfaces. **Figure 1a** illustrates the chemical structure (top) and the corresponding electrostatic potential map (bottom) of the GA<sup>+</sup> molecule. The molecule has multiple functional groups, including electron-rich amine and imine and electron-deficient ammonium groups. Negatively charged iodide ions compensate for the positive charge on the ammonium head group. Due to the different charged environments of functional groups, the electron cloud around the molecule has electron-rich and electron-deficient domains, which are highly beneficial for passivating defects on complex perovskite surfaces. To support this, DFT calculations were performed on supercells with a composition closely resembling FAPbI<sub>3</sub> (with 0.8 mol% MAPbBr<sub>3</sub>) used in the Experimental Section of this study. Three different perovskite surfaces were constructed: formamidinium (FA)-deficient surface (lacking FA atoms), undercoordinated iodide (I) surface (i.e., Pb-deficient surface), and undercoordinated lead (Pb) surface (i.e., I-deficient surface), as shown in **Figure 1b–d**. These surfaces are denoted as



**Figure 2.** a) Pb 4f, b) I 3d, c) N 1s XPS spectra of the perovskite films without and with GAI modification. d) FTIR spectra of pure GAI and GAI mixed with PbI<sub>2</sub>. e) The schematic mechanism of GAI modulation on perovskite.

FA-deficient, Pb-deficient, and I-deficient surfaces, respectively. The obtained electron density profiles are shown below the surfaces. These profiles emphasize that the perovskite surfaces consist of a mixture of electron-rich and electron-deficient domains greatly influenced by surface defects. Therefore, multi-functional guanidinium iodide molecules are expected to effectively passivate various surface defects by combining electron-rich and electron-deficient groups.

Next, we investigated the interaction between GA<sup>+</sup> molecules and the three surfaces. Table S1 (Supporting Information) summarizes the obtained binding energies and Bader charges. The Pb-deficient surface (−5.943 eV) achieved the highest binding energy, followed by the FA-deficient (−5.437 eV) and I-deficient (−2.947 eV) surfaces. These values are also reflected in the Bader charges, where positive values indicate charge transfer from the molecule to the surface. Moreover, the electron density difference was calculated to understand the nature of bonding between the molecule and the surface (Figure 1e–g). The turquoise and yellow regions represent charge depletion and accumulation, respec-

tively. It is evident that GA<sup>+</sup> can reorient itself based on the nature of surface defects, thus capable of passivating various types of surface dangling bonds in perovskites. Its electron-deficient ammonium groups (NH<sup>3+</sup>) primarily interact with negatively charged defects resulting from A-site cation vacancies, followed by interactions with electronegative surface I atoms. Conversely, unsaturated nitrogen (N) atoms in electron-rich amines and imines primarily interact with electron-deficient Pb surface atoms, forming the strongest bonds. The electron density difference indicates that a GA<sup>+</sup> molecule can simultaneously passivate various surface dangling bonds.

We performed X-ray photoelectron spectroscopy (XPS) analysis on the thin films to further elucidate the interaction mechanism between GAI and perovskite. As shown in Figure 2a, the Pb 4f XPS spectra reveal that the Pb 4f<sub>5/2</sub> and 4f<sub>7/2</sub> peaks for the control perovskite thin films are at 143.22 and 138.32 eV, respectively. Following GAI treatment, these peaks exhibit a shift to lower binding energies. This shift is ascribed to the interaction between the lone-pair electrons of the unsaturated amine



(or imine) groups in the  $\text{GA}^+$  cation and the undercoordinated  $\text{Pb}^{2+}$  ions, increasing electron density around Pb.<sup>[19,20]</sup> In a similar vein, the  $\text{I } 3\text{d}^{3/2}$  and  $\text{I } 3\text{d}^{5/2}$  peaks, initially positioned at 630.61 and 619.15 eV, respectively, are observed to shift to 630.47 and 619.03 eV upon the introduction of GAI (refer to Figure 2b). This indicates that the iodide ions from GAI can effectively compensate for halide vacancies within the perovskite matrix.<sup>[21]</sup> Moreover, a subtle shift in the N1s peak is detected (as depicted in Figure 2c), likely influenced by the ammonium groups in GA. This observation confirms the presence of  $\text{GA}^+$  cations on the surface of the treated perovskite, contributing to defect passivation by providing additional ammonium groups.<sup>[22,23]</sup> The deconvolution of N1s XPS spectra is shown in Figure S1 (Supporting Information). The binding energy of C–N shifts from 400.16 to 399.95 eV, while that of C=N shifts from 401.02 to 400.73 eV. Notably, once GAI is introduced, the GAI, which consists of additional ammonium groups, remains on the top surface and reacts with the perovskite film. Fourier-transform infrared spectroscopy (FTIR) in Figure 2d shows that the stretching vibration of C=N shifted from  $1654.63 \text{ cm}^{-1}$  in pure GAI to a lower wavenumber at  $1644.98 \text{ cm}^{-1}$  in GAI mixed with  $\text{PbI}_2$  as the empty orbitals of  $\text{Pb}^{2+}$  could receive unpaired lone electrons from C=N, suggesting intensive chemical interactions between GAI and  $\text{Pb}^{2+}$ .<sup>[24]</sup>

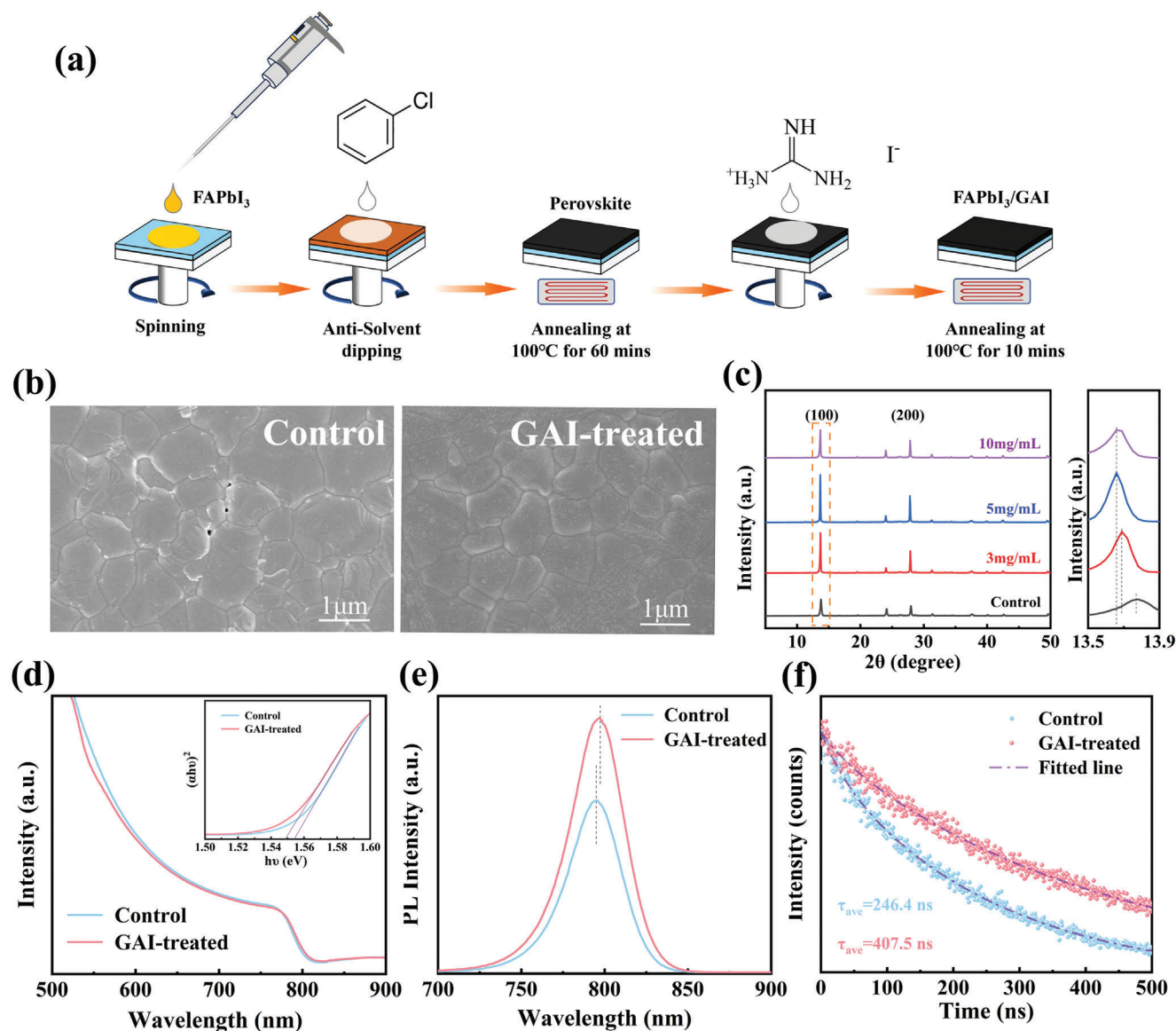
According to DFT and experimental studies, we propose a model to elucidate the modulation mechanism of GAI on perovskite, as depicted in Figure 2e. The  $\text{GA}^+$  features two distinct nitrogen environments: the positively charged  $-\text{NH}_3^+$  and two unsaturated nitrogen sites, which could be amines ( $-\text{NH}_2$ ) or imines ( $=\text{NH}$ ). The ammonium cation neutralizes negatively charged defects arising from the lack of A-site atoms through ionic interactions, while the amine or imine group, with its lone pair of electrons, serves as a Lewis base to anchor uncoordinated  $\text{Pb}^{2+}$  defects by donating these electrons.<sup>[25]</sup> Additionally, the iodide ions present in GAI can compensate for iodine deficiencies that may occur during perovskite film formation, thus passivating iodine vacancy defects. As a result, GAI treatment effectively rectifies various intrinsic defects across multiple reactive sites within the perovskite structure, and devices treated with GAI are expected to demonstrate improved photovoltaic performance.

**Figure 3a** outlines the standard one-step anti-solvent method used to fabricate FA-based perovskite thin films. After this process, GAI solutions with varying concentrations were spin-coated onto the films and annealed at  $100^\circ\text{C}$  for 10 min. Scanning electron microscopy (SEM) examined the evolution of the surface morphology, as shown in Figure 3b. The control perovskite film displayed significant pinholes, while the film treated with a  $3 \text{ mg mL}^{-1}$  GAI solution exhibited complete surface coverage, characterized by a more uniform crystal texture. The surface morphology of GAI-treated perovskite film was also examined using atomic force microscopy (AFM), see Figure S2 (Supporting Information); the optimized film appears smoother than the control perovskite film. The GAI-treated films did not exhibit the white edges observed at grain boundaries in the control films, suggesting improved charge transport properties in the treated samples. The smoother surface and refined grain boundaries facilitate better interfacial contact, enhancing the extraction and transport efficiency of charge carriers in GAI-treated devices.<sup>[26]</sup> These observed variations in film morphology highlight the beneficial ef-

fects of GAI treatment on defect passivation within the perovskite thin films. It should be noted that there are uneven distributions resembling dendritic structures on the surface of the perovskite films as the GAI concentration gets higher, as shown in Figure S3 (Supporting Information).

Structural characterization of the perovskite films was performed using X-ray diffraction (XRD) (Figure 3c). Notably, strong diffraction peaks corresponding to the (100) and (200) crystal planes of the perovskite films are observed at  $\approx 14^\circ$  and  $28^\circ$ , respectively. Compared to the control perovskite, films treated with 3 and  $5 \text{ mg mL}^{-1}$  GAI in an isopropanol solution show increased XRD peak intensity and a reduced full width at half-maximum (FWHM). These observations indicate improved crystallinity and higher quality in the GAI-treated perovskite films. As the GAI concentration increases, the (100) peaks are observed to shift toward smaller diffraction angles (denoted as  $2\theta$ ). This shift suggests a lattice expansion attributed to incorporating the larger guanidinium cations into the perovskite lattice.<sup>[27]</sup> This phenomenon confirms the integration of these cations into the crystal structure. Figure 3d displays the UV–vis absorption spectra of perovskite films, revealing a subtle redshift in the absorption spectra of GAI-treated perovskite films. Tauc analysis, as illustrated in the inset of Figure 3d, suggests that GAI treatment results in a marginal reduction in the perovskite bandgap. As detailed in the experimental section, Methylamine ions ( $\text{MA}^+$ ) are incorporated into the  $\text{FAPbI}_3$  lattice as additives. The incorporation of GA is posited to mitigate the local lattice strain induced by the smaller  $\text{MA}^+$ , consequently leading to a slight decrease in the bandgap.<sup>[28,29]</sup> This finding aligns with the XRD diffraction patterns presented in Figure 3c, further corroborating the integration of guanidinium cations into the perovskite lattice. Consequently, the introduction of larger cations can alleviate lattice compression strain, thereby reducing defect centers that trap charge carriers and enhancing the intrinsic stability of the perovskite thin films.<sup>[28]</sup> The variation in lattice strain is also related to the Urbach energy ( $E_U$ ).<sup>[30]</sup> Figure S4 (Supporting Information) depicts the Urbach energy of the perovskite thin films derived from the UV–vis absorption spectra. The  $E_U$  values for the control and GAI-treated perovskite films are 31.34 and 30.53 meV, respectively. The lower  $E_U$  observed in the GAI-modified perovskite film indicates superior structural quality. A reduced  $E_U$  value also signifies excellent electronic properties, such as enhanced carrier mobility.

Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements were utilized to evaluate the effects of GAI on the optoelectronic characteristics of perovskite films. As depicted in Figure 3e, the perovskite thin films treated with GAI display increased fluorescence intensity, which suggests that GAI treatment effectively mitigates non-radiative recombination resulting from defects within the perovskite film.<sup>[31,32]</sup> This outcome is consistent with the enhanced film morphology and elevated crystallinity observed through SEM and XRD analyses of the GAI-treated films. Additionally, a minor redshift in the PL peak position is detected for the GAI-treated perovskite, which may be ascribed to the introduction of gap states induced by the GAI treatment.<sup>[33]</sup> This agrees with the slightly reduced optical bandgap noted in Figure 3d. Figure 3f presents the TRPL decay curves for both the control and GAI-



**Figure 3.** a) Flowchart of perovskite film fabrication process. b) Top-view SEM images of control and 3 mg mL<sup>-1</sup> GAI-treated perovskite films. c) X-ray diffraction (XRD) patterns of perovskite thin films treated with various concentrations of GAI, along with a local enlargement of the (100) peak. d) UV-vis absorption and Tauc plot of perovskite films without and with GAI modification. e) steady-state photoluminescence (PL) spectra. and f) time-resolved photoluminescence (TRPL) spectra of the perovskite films.

treated perovskite films. The TRPL decay curves were fitted using a bi-exponential model expressed as follows:<sup>[34]</sup>

$$I(t) = I_0 + A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) \quad (1)$$

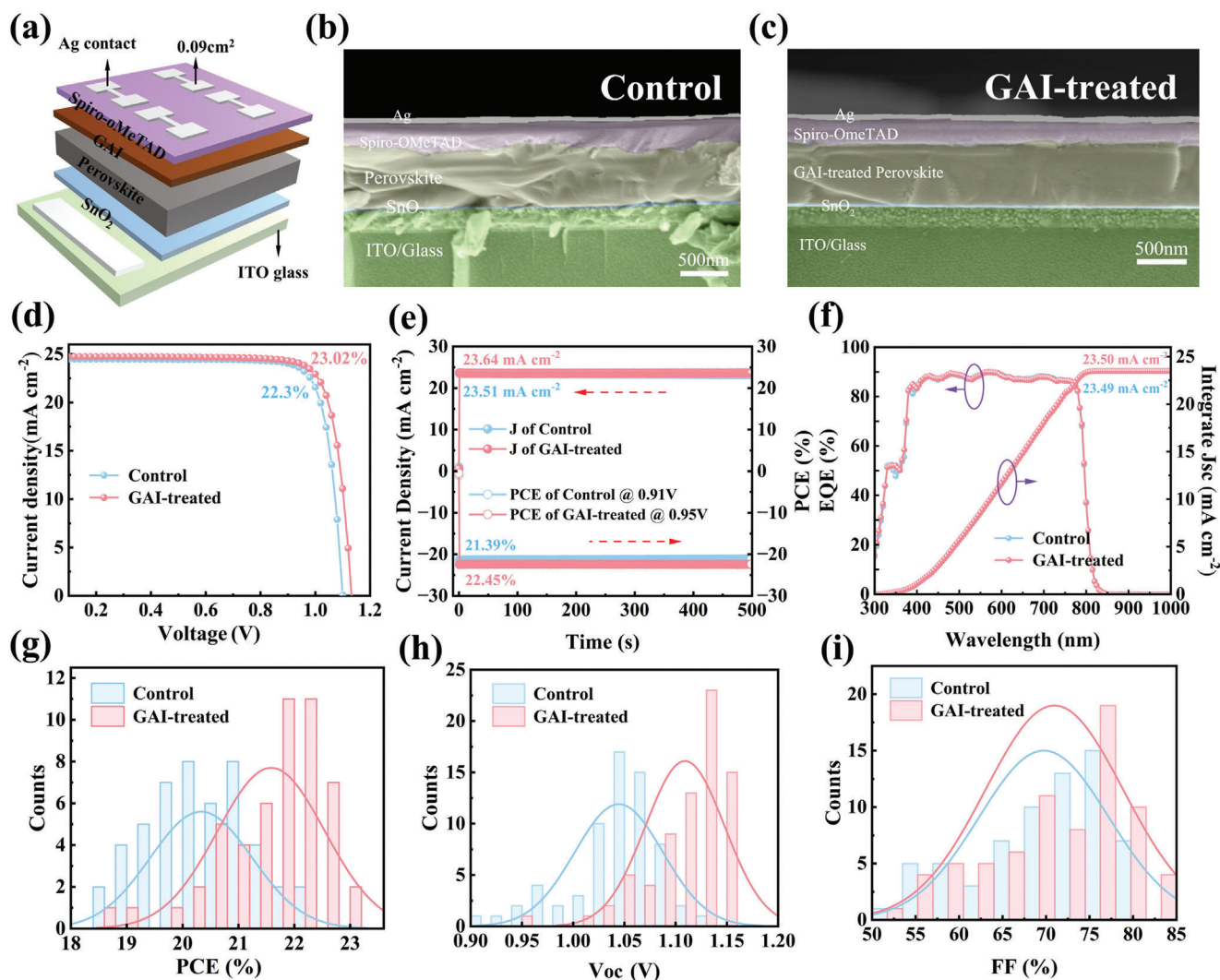
The average lifetime constant ( $\tau_{ave}$ ) was calculated using the equation:<sup>[35]</sup>

$$\tau_{ave} = \frac{\sum A_i \tau_i^2}{\sum A_i \tau_i} \quad (2)$$

where  $\tau_1$  and  $\tau_2$  denote the fast and slow decay lifetimes, respectively. The fitting parameters and calculated  $\tau_{ave}$  values are compiled in Table S2 (Supporting Information). The GAI-treated per-

ovskite film shows a significantly longer lifetime ( $\tau_{ave} = 407.5$  ns) than the control film ( $\tau_{ave} = 246.4$  ns), nearly double. This extended lifetime indicates a reduced defect-related carrier recombination in the GAI-treated perovskite.<sup>[36]</sup>

Solar cells featuring an n-i-p structure were fabricated to evaluate the effects of GAI treatment on device performance. The solar cell structure comprises tin oxide (SnO<sub>2</sub>) as the electron transport layer and 2,2',7,7'-tetrakis(N,N-di-p-methoxyphenylamine)-9,9'-spirobifluorene (Spiro-OMeTAD) as the hole transport layer, as depicted in the inset of Figure 4a. The cross-sectional SEM images of PSCs in Figure 4b,c reveal that devices treated with GAI exhibit smoother interfaces between the perovskite layer and the hole transport layer. The enhancement in interface quality is



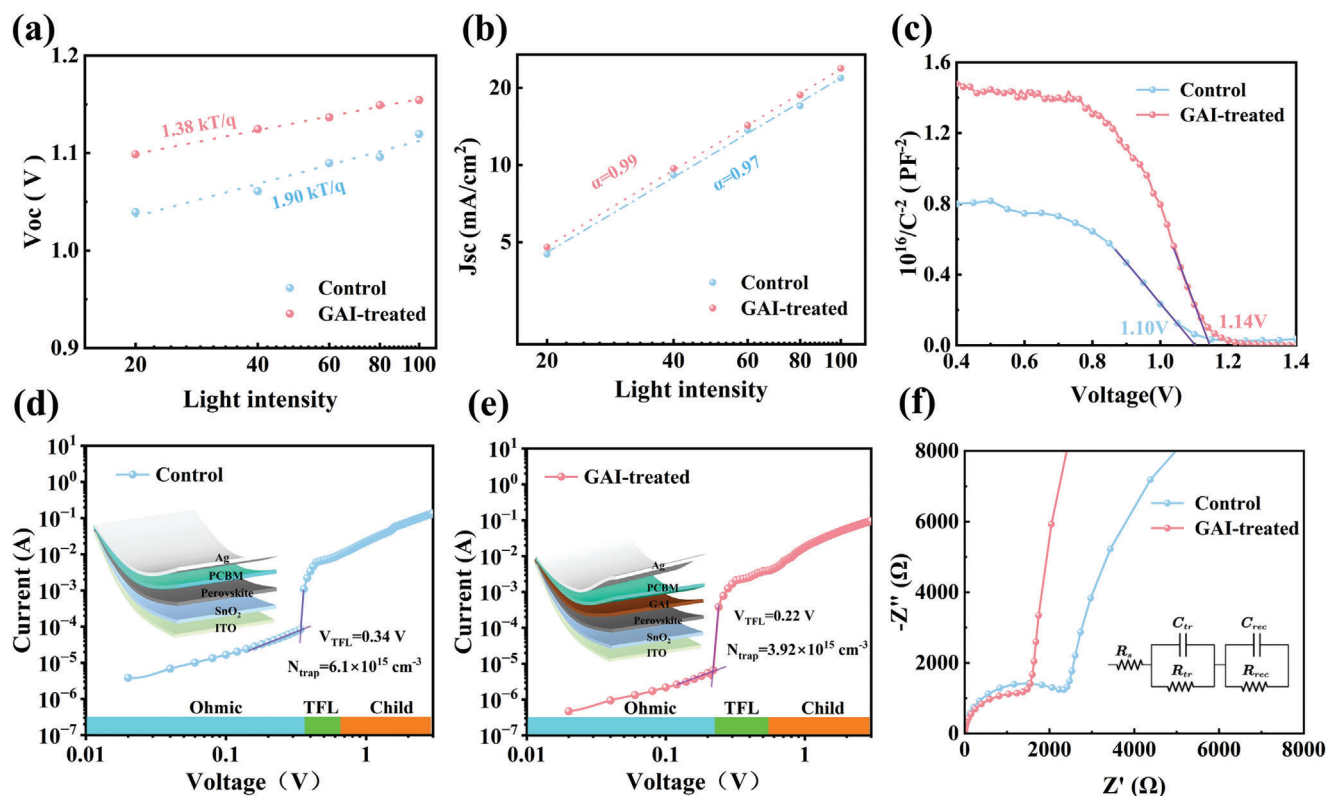
**Figure 4.** a) Illustration of the n-i-p device structure used in this study. Cross-sectional SEM images of b) Control cell and c) GAI-treated cell. d) J-V curves of the champion device with/without GAI-treated. e) Steady-state power output and steady-state photocurrent near MPP for the champion devices. f) EQE and integrated  $J_{SC}$  of control and GAI devices. Statistics of photovoltaic parameters g) PCE, h)  $V_{OC}$ , and i) FF of control and 3 mg mL<sup>-1</sup> GAI devices.

attributed to the reaction between GA<sup>+</sup> and uncoordinated Pb and/or Pb clusters on the perovskite surface. Benefiting from its favorable penetration characteristics within the perovskite film, GAI can permeate throughout the bulk of the perovskite, enabling modulation of morphology both within the bulk and at the interface. This improved morphology provides the potential for enhanced charge extraction and suppression of non-radiative recombination, thereby contributing to improved device performance. Figure S5 (Supporting Information) provides statistical data on the performance of solar cell devices treated with varying concentrations of GAI, revealing that devices treated with 3 mg/mL GAI achieve the highest PCE. The efficiency of the GAI-treated device increased from 22.3% to 23.02% compared to the control, as presented in Figure 4d. The stabilized power output (SPO) and steady-state photocurrent curves were measured during maximum power point (MPP) tracking for 500 s to confirm the reliability of the photovoltaic performance during operation (Figure 4e). The GAI-treated device exhibited an SPO 22.45%

higher than that of the control device, which is consistent with the PCE obtained from the J-V test. The external quantum efficiency (EQE) curve of the devices is shown in Figure 4f, with the integrated current aligning with the short-circuit current density ( $J_{sc}$ ) of the J-V curve. The photovoltaic parameters of 60 individual control and GAI-treated cells were statistically analyzed. Figure 4g highlights a notable improvement in efficiency and reproducibility for the GAI-treated devices. Figure 4h,i illustrates that the primary contributors to the enhanced device efficiency are a significant increase in the open-circuit voltage ( $V_{OC}$ ) and an improved fill factor (FF). Additionally, the PCE of devices with higher GAI concentrations deteriorates due to an uneven interface between the perovskite film and the hole transport layer, as depicted in Figure S3 (Supporting Information), leading to reduced open-circuit voltage and fill factor.

GAI-treated devices achieve higher PCE, primarily due to a substantial increase in  $V_{OC}$ . The relationship between light intensity and  $V_{OC}$  is utilized to evaluate the diode characteristics of





**Figure 5.** The influence of light intensity on a)  $V_{OC}$  and b)  $J_{SC}$  of PSCs with and without  $3 \text{ mg mL}^{-1}$  GAI. c) Mott–Schottky curves. SCLC characteristics of electron-only devices for d) control and e)  $3 \text{ mg mL}^{-1}$  GAI. f) Nyquist plots of the devices at  $V = 0.4 \text{ V}$ .

perovskite solar cells.  $V_{OC}$  can be derived by setting the current to zero in the Shockley equation, which is expressed as follows:

$$V_{OC} = nkTq^{-1} \ln(J_{sc}/J_0 + 1) \quad (3)$$

where  $n$ ,  $k$ ,  $T$ ,  $q$ , and  $J_0$  represent the ideality factor, Boltzmann constant, device temperature, elementary charge, and dark saturation current density, respectively. Since  $J_{SC}$  is directly proportional to the light intensity and significantly exceeds  $J_0$ ,  $V_{OC}$  is proportional to the logarithm of the light intensity with a slope of  $nkTq^{-1}$ . Typically,  $n$  is equal to 1 without trap-assisted carrier recombination (monomolecular SRH recombination) in the devices.<sup>[37]</sup> As depicted in **Figure 5a**, the control device exhibits an ideality factor of 1.90, whereas the GAI-treated device shows a reduced ideality factor of 1.38. This decrease in the ideality factor suggests that GAI treatment suppresses trap-assisted non-radiative recombination at the interface, leading to increased  $V_{OC}$ . Furthermore, the double logarithmic relationship between  $J_{SC}$  and light intensity is calculated based on the proportional relationship between the light intensity power exponent and  $J_{SC}$ .<sup>[38]</sup> An ideal device, free from bi-molecular recombination involving free carriers, should exhibit an  $\alpha$  value close to unity.<sup>[39]</sup> The GAI-treated device achieves a larger  $\alpha$  value of 0.99 compared to the control device's 0.97, approaching unity, as shown in **Figure 5b**. This indicates that adding GAI reduces deep trap states by passivating grain boundaries, allowing for more effective extraction of charge carriers. These findings are consistent with the TRPL analysis of perovskite films.

To gain a more profound comprehension of the factors contributing to the enhancement of  $V_{OC}$ , as illustrated in **Figure 5c**, the built-in potential ( $V_{bi}$ ) of the device is calculated using the Mott–Schottky (M-S) equation:

$$\frac{1}{C^2} = \frac{2}{A^2 \epsilon q} \cdot \frac{1}{N_A} \cdot (V_{bi} - V) \quad (4)$$

where  $A$  is cell area,  $\epsilon$  is the dielectric constant of perovskite,  $q$  is the elementary charge,  $N_A$  is doping density. The calculated  $V_{bi}$  for the control and GAI-treated devices are 1.07 and 1.16 V, respectively. The increased  $V_{bi}$  demonstrates that build-in band alignment contributes a stronger driving force for the intrinsic charge separation. Additionally, the carrier density at interface is inversely proportional to the slope value of the linear regime. The lower carrier density of GAI-treated device testifies the faster and more effective charge transfer at the perovskite/HTL interface, which is responsible for the enhancement of  $V_{OC}$  as well.<sup>[40]</sup>

To further explore the effect of GAI modification on defect passivation at device interfaces, **Figure 5d,e** illustrate the use of the space-charge limited current (SCLC) method to assess the trap density in perovskite with the structure of ITO/SnO<sub>2</sub>/Perovskite/PCBM/Ag. The trap density ( $n_{trap}$ ) is calculated from the trap filling limit voltage ( $V_{TFL}$ ) using the equation:

$$n_{trap} = \frac{2\epsilon V_{TFL}}{qL^2} \quad (5)$$

where  $\epsilon$  is the dielectric constant of perovskite (46.9),<sup>[41,42]</sup>  $q$  is the elementary charge,  $L$  is the thickness of perovskite film.<sup>[43]</sup> The calculated electron trap density for the GAI-treated device is  $3.92 \times 10^{15} \text{ cm}^{-3}$ , which is lower than the  $6.1 \times 10^{15} \text{ cm}^{-3}$  observed for the control device. This reduction in defect density is attributed to the passivation of under-coordinated  $\text{Pb}^{2+}$  at the grain boundaries by  $\text{GA}^+$  and the suppression of iodine vacancy formation.<sup>[17]</sup> This suggests that the GAI treatment significantly reduces the trap density within the device, enhances charge separation, and improves charge carrier dynamics, ultimately boosting PSCs' open-circuit voltage ( $V_{OC}$ ) and fill factor (FF), thereby improving their overall performance. Consequently, we have calculated the carrier mobility of the GAI-modified perovskite layer using the SCLC method. The carrier mobility is determined according to Mott–Gurney's law:

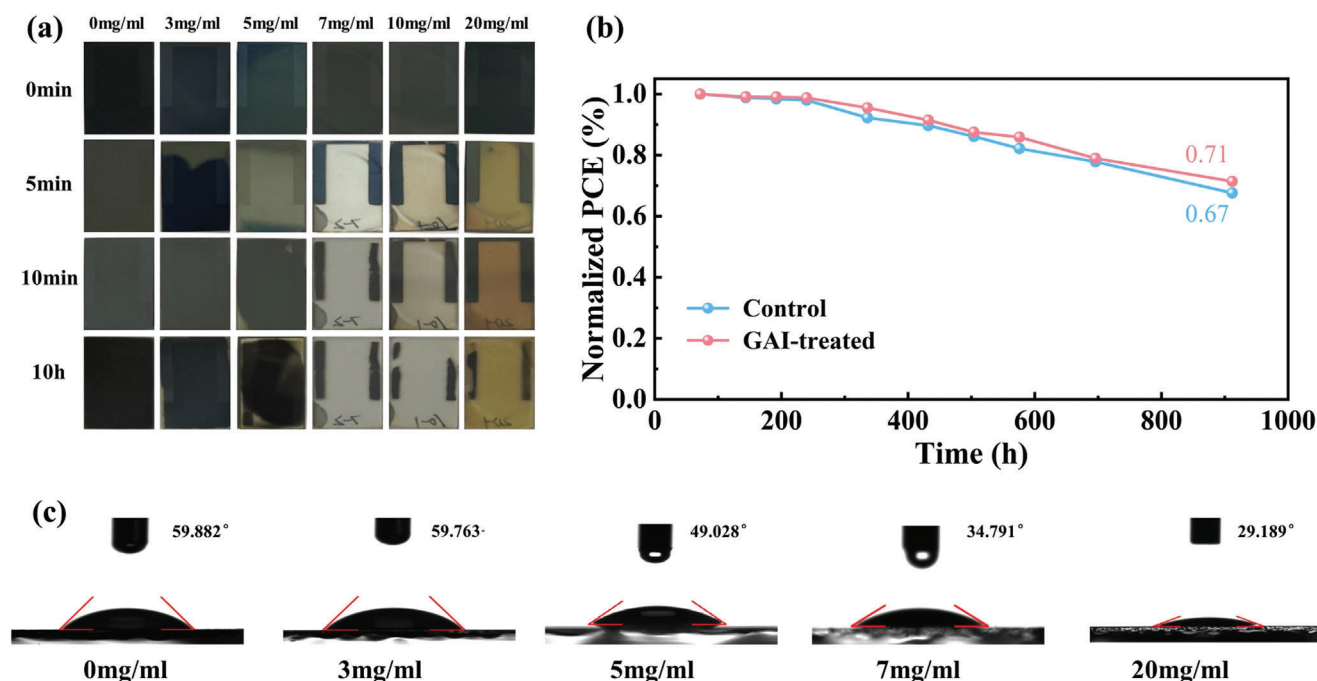
$$\mu = \frac{8J_D L^3}{9\epsilon_r \epsilon_0 V^2} \quad (6)$$

where  $J_D$  is the current density.<sup>[44]</sup> As indicated in Table S3 (Supporting Information), the GAI-modified perovskite film demonstrates superior carrier transport capabilities compared to the control group.

Electrochemical impedance spectroscopy (EIS) is a widely used characterization method in the frequency domain that enables the decoupling of processes associated with different characteristic time scales, and it has been employed to study PSCs since the early stages of their research. In this context, EIS spectra are utilized to elucidate the interfacial charge transfer kinetics of PSCs. Nyquist plots, measured at a bias voltage of 0.4 V under dark conditions, are displayed in Figure 5f, revealing two distinct arcs in both the high-frequency and low-frequency regions. The

equivalent circuit model depicted in the inset of Figure 5f comprises a series combination of two elements: a resistor and capacitor in parallel ( $R_{tr}$  and  $C_{tr}$ ) and a series resistor ( $R_s$ ). The corresponding values are compiled in Table S4 (Supporting Information). The  $R_s$  values for both control and treated devices are comparable and relatively low, indicating that the series resistance contributed by the electrodes and connecting wires can be neglected in the analysis of interfacial charge kinetics. The  $R_{tr}$ , corresponding to the high-frequency arc, reflects the charge transfer characteristics at the interface. The lower  $R_{tr}$  of the treated devices suggests more efficient charge extraction and transfer. The  $R_{rec}$ , associated with the low-frequency arc, indicates the interfacial recombination kinetics. The higher  $R_{rec}$  of the treated devices suggests a suppression of charge recombination at the interface.<sup>[45]</sup> These observations can be attributed to the reduced migration of iodide ions at the interface and the effective passivation of interface defects following GAI treatment. Additionally, Table S5 (Supporting Information) shows that the GAI-treated perovskite solar cell exhibits reduced hysteresis. Generally, the hysteresis of PSCs can be attributed to the interfacial capacitance caused by charge accumulation at the interface, which originates from high trap density and unbalanced charge transport within perovskite devices.<sup>[46]</sup> The GAI-treated PSCs have lower trap density and higher carrier mobility, which reduces charge accumulation at the perovskite interface and suppresses the hysteresis effect of PSCs.

To assess the stability of the solar cells, we initially examined the perovskite films' stability when stored in an environment maintained at 25 °C with a relative humidity above 50%. Figure 6a displays photographs of the control device and perovskite films treated with various concentrations of GAI after



**Figure 6.** a) Photographs of the control devices and perovskite films treated with varying concentrations of GAI after being exposed to air at 25 °C with humidity above 50% for different durations. b) Normalized PCE variation curves of the devices after storing under ambient conditions with 30–40% RH at room temperature. c) The contact angles of perovskite films treated with varying concentrations of GAI.



exposure to air for different periods. It was noted that high concentrations of GAI treatment did not enhance device efficiency (as depicted in Figure S5, Supporting Information) and, conversely, resulted in accelerated degradation of the films. Figure 6c presents the water contact angles of the perovskite films following different GAI treatments, indicating a slight decrease in the contact angle with an increase in GAI concentration. This decrease is attributed to GAI's hydrophilic nature; the  $\text{GA}^+$  contains amine groups prone to absorbing moisture and forming hydrogen bonds.<sup>[47]</sup> Consequently, films treated with GAI exhibit greater hydrophilicity compared to untreated films. At lower GAI concentrations, GAI infiltrates the perovskite lattice, forming a passivation layer on the surface with minimal impact on hysteresis. In contrast, higher GAI concentrations lead to the formation of a GAI-rich layer on the perovskite surface, which is highly hydrophilic and accelerates film degradation. We further explored the stability of perovskite solar cells treated with 3 mg mL<sup>-1</sup> GAI in a dark environment maintained at 25 °C with a relative humidity of 30% to 40%, as illustrated in Figure 6b. Figure S6 (Supporting Information) shows the thermal stability of the unencapsulated device under 65 °C sustaining hot pressing in an  $\text{N}_2$ -filled glove box. These results unambiguously show that the GAI-treated PSCs have similar device stability to the control devices. This suggests that optimizing the amount of GAI can weaken its hydrophilic nature and balance the efficiency and stability of the PSC devices.

### 3. Conclusion

The interface is critical to efficient perovskite solar cells, and the poor top interface could lead to defects and recombination, which limit the device's performance and stability. We have fabricated FA-based perovskite films with enhanced surface coverage and grain boundary quality by applying GAI surface passivation. The optimal concentration of GAI markedly diminishes the grain boundary defects, improves crystallinity, and avoids the adverse effects of its hydrophilic properties. We suggest that the synergistic action of the ammonium group, unsaturated nitrogen atoms, and iodide ions within GAI can effectively repair surface defects of the perovskite via multiple pathways, thereby significantly suppressing trap-assisted non-radiative recombination. This enhancement is directly manifested in the improved PCE from 22.3% to 23.02%. This highlights the critical role of accurately controlling GAI application during passivation. Our research illustrates that the judicious use of GAI passivation is a swift and potent approach for fabricating high-performance perovskite solar cells.

### 4. Experimental Section

**Materials:**  $\text{PbI}_2$  (99.99%), formamidinium iodide (FAI, 99.5%), methylammonium chloride (MACl, 99.5%), methylammonium bromide (MABr), Lead iodide ( $\text{PbI}_2$ ) lead bromide ( $\text{PbBr}_2$ ), guanidinium iodide (GAI, 99.9%), N,N-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.8%), chlorobenzene (CB, 99.9%), 4-tertbutylpyridine (tBP, >99%), acetonitrile (ACN, 99.9%), lithium salt-bis (trifluoromethane) sulfonimide imide (LiTFSI), 2,2',7,7'-tetrakis-(N,N-dimethoxyphenylamine)-9,9'-spirobifluorene (spiro-OMeTAD, 99.86%), and indium tin oxide (ITO) were purchased from Yingkou

Advanced Election Technology Co., Ltd.  $\text{SnO}_2$  colloidal solution (12 wt% in  $\text{H}_2\text{O}$ ) was purchased from Xian Yuri Solar Co., Ltd. FK269 Co(III) TFSI salt (>99%) was purchased from Shanghai Zaofu New Materials Technology Co., Ltd. Isopropanol (IPA, >95%) was purchased from Shanghai Lingfeng Chemical Reagent Co., Ltd. Ethanol (>95%) was purchased from Sinopharm Chemical Reagent Co., Ltd.

**Device fabrication:** ITO was sequentially cleaned with deionized water, IPA, and ethanol for 15 min, respectively. Before use, the ITO/glass substrates were treated with ultraviolet ozone for 15 min. Then, the ITO/glass substrates were spin-coated with a thin layer of  $\text{SnO}_2$ , diluted in water at a 1:3 ratio, at 3000 rpm for 45 s (500 rpm ramp), followed by annealing at 150 °C for 30 min. Before perovskite deposition, the ITO/ $\text{SnO}_2$  substrate was oxygen plasma treated and pumped inside a nitrogen glovebox. A FA-based perovskite composition with excess  $\text{PbI}_2$  stoichiometry (9 mol%) and MACl as an additive was used. The perovskite solution was prepared by mixing 1.53 M  $\text{PbI}_2$ , 1.4 M FAI, 0.5 M MACl, 0.0122 M MABr, and 0.0122 M  $\text{PbBr}_2$  in DMF:DMSO = 8:1. The perovskite precursor was left to stir overnight until fully dissolved. The perovskite solution deposited via spin coating at 1000 rpm for 10 s, and 5000 rpm for 30 s (both 2000 rpm ramp). One hundred twenty microliters of chlorobenzene (CBZ) was poured onto the substrate 15 s prior to the end of the second step. Afterward, the ITO/ $\text{SnO}_2$ /perovskite substrate was annealed at 100 °C for 60 min. For GAI treatment, a certain concentration of GAI in IPA solution was spin-coated onto the perovskite film at 5000 rpm for 30 s with subsequent annealing for 10 min at 100 °C. The hole transporting layer (HTL) was deposited by preparing the HTL solution consisting of 72.3 mg of Spiro-OMeTAD, 28.8  $\mu\text{L}$  of tBP, 10  $\mu\text{L}$  of FK269 Co(III) TFSI solution (300 mg in 1 mL acetonitrile), 17.5  $\mu\text{L}$  of Li-TFSI solution (520 mg in 1 mL acetonitrile), and 1 mL of chlorobenzene. Twenty-five microliters of the HTL solution was loaded onto the perovskite substrate and spin-coated at 4000 rpm for 30 s with a 3000 rpm s<sup>-1</sup> ramp. Preparation and deposition of the HTL solution were performed inside a nitrogen glovebox. The Ag electrode (100 nm) was deposited by thermal evaporation.

**Characterizations:** UV-vis: The UV-2600 UV-vis spectrophotometer from Shimadzu Corporation was used to measure the optical absorption and transmission characteristics of the thin films. PL/TRPL: Steady-state photoluminescence (PL, with an excitation wavelength of 445 nm) and time-resolved photoluminescence (TRPL) were measured using an FLS1000 spectrometer (Edinburgh Instruments Ltd.). The samples consisted of films coated on plain glass substrates, with excitation and emission PL measured on the side of the substrate where the film was applied. XRD: X-ray diffraction (XRD) patterns were acquired by a D/MAX Ultima III diffractometer (Rigaku Corporation, Japan). J-V: The J-V curves of the PSC devices were measured by a Keithley 2400 source meter with a delay of 1 ms in a room environment. The light source was a solar simulator (Newport, 81904, 3 A) at AM 1.5G (100 mW cm<sup>-2</sup>) calibrated by a standard Si cell. The active area of the PSC devices was 0.9 cm<sup>2</sup> with a shadow mask. Contact angle measurements: Contact angle measurements were conducted using the FCA2000A from Shanghai Aifisi Precision Instruments Co., Ltd., equipped with high-quality optical components to ensure high precision in displaying the droplet, thereby accurately measuring the contact angle. The measurements were performed with 2  $\mu\text{L}$  of deionized water.

### Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Keywords

guanidinium, interface passivation, non-radiative recombination, perovskites

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- [1] A. Kojima, K. Teshima, Y. Shirai, T. Miyasaka, *J. Am. Chem. Soc.* **2009**, *131*, 6050.
- [2] Y. Hou, E. Aydin, M. De Bastiani, C. Xiao, F. H. Isikgor, D.-J. Xue, B. Chen, H. Chen, B. Bahrami, A. H. Chowdhury, A. Johnston, S.-W. Baek, Z. Huang, M. Wei, Y. Dong, J. Troughton, R. Jalmood, A. J. Mirabelli, T. G. Allen, E. Van Kerschaver, M. I. Saidaminov, D. Baran, Q. Qiao, K. Zhu, S. De Wolf, E. H. Sargent, *Science* **2020**, *367*, 1135.
- [3] H. Chen, F. Ye, W. Tang, J. He, M. Yin, Y. Wang, F. Xie, E. Bi, X. Yang, M. Grätzel, L. Han, *Nature* **2017**, *550*, 92.
- [4] G. E. Eperon, S. D. Stranks, C. Menelaou, M. B. Johnston, L. M. Herz, H. J. Snaith, *Energy Environ. Sci.* **2014**, *7*, 982.
- [5] NREL, Best Research-Cell Efficiencies, <https://www.nrel.gov/pv/assets/pdfs/best-research-cell-efficiencies.pdf>, (accessed: March 2024).
- [6] P. Wu, S. Wang, X. Li, F. Zhang, *Matter* **2022**, *5*, 1137.
- [7] F. Zhang, K. Zhu, *Adv. Energy Mater.* **2020**, *10*, 1902579.
- [8] T. A. S. Doherty, A. J. Winchester, S. Macpherson, D. N. Johnstone, V. Pareek, E. M. Tennyson, S. Kosar, F. U. Kosasih, M. Anaya, M. Abd-Jalebi, Z. Andaji-Garmaroudi, E. L. Wong, J. Madéo, Y.-H. Chiang, J.-S. Park, Y.-K. Jung, C. E. Petoukhoff, G. Divitini, M. K. L. Man, C. Ducati, A. Walsh, P. A. Midgley, K. M. Dani, S. D. Stranks, *Nature* **2020**, *580*, 360.
- [9] J. Chen, N.-G. Park, *ACS Energy Lett.* **2020**, *5*, 2742.
- [10] J. Chen, N. Park, *Adv. Mater.* **2019**, *31*, 1803019.
- [11] J. He, J. He, D. Ma, J. Sheng, W. Shao, T. Ding, W. Wu, *Small* **2024**, *20*, 2305127.
- [12] F. Qiu, Q. Liu, Y. Liu, J. Wu, *Small* **2023**, *19*, 2304834.
- [13] Y. Cho, A. M. Soufiani, J. S. Yun, J. Kim, D. S. Lee, J. Seidel, X. Deng, M. A. Green, S. Huang, A. W. Y. Ho-Baillie, *Adv. Energy Mater.* **2018**, *8*, 1703392.
- [14] Y. Zhou, H. Xue, Y. Jia, G. Brocks, S. Tao, N. Zhao, *Adv. Funct. Mater.* **2019**, *29*, 1905739.
- [15] F. Zhang, B. Cai, J. Song, B. Han, B. Zhang, H. Zeng, *Adv. Funct. Mater.* **2020**, *30*, 2001732.
- [16] W. Li, Z. Wang, F. Deschler, S. Gao, R. H. Friend, A. K. Cheetham, *Nat. Rev. Mater.* **2017**, *2*, 16099.
- [17] N. De Marco, H. Zhou, Q. Chen, P. Sun, Z. Liu, L. Meng, E.-P. Yao, Y. Liu, A. Schiffer, Y. Yang, *Nano Lett.* **2016**, *16*, 1009.
- [18] N. D. Pham, C. Zhang, V. T. Tiong, S. Zhang, G. Will, A. Bou, J. Bisquert, P. E. Shaw, A. Du, G. J. Wilson, H. Wang, *Adv. Funct. Mater.* **2019**, *29*, 1806479.
- [19] D.-H. Kang, S.-Y. Kim, J.-W. Lee, N.-G. Park, *J. Mater. Chem.* **2021**, *9*, 3441.
- [20] H. Hsu, H. Hsiao, T. Juang, B. Jiang, S. Chen, R. Jeng, C. Chen, *Adv. Energy Mater.* **2018**, *8*, 1802323.
- [21] Q. Jiang, Y. Zhao, X. Zhang, X. Yang, Y. Chen, Z. Chu, Q. Ye, X. Li, Z. Yin, J. You, *Nat. Photonics* **2019**, *13*, 460.
- [22] P. Wu, S. Wang, J. H. Heo, H. Liu, X. Chen, X. Li, F. Zhang, *Nano-Micro Lett.* **2023**, *15*, 114.
- [23] K. Wang, S. Ma, X. Xue, T. Li, S. Sha, X. Ren, J. Zhang, H. Lu, J. Ma, S. Guo, Y. Liu, J. Feng, A. Najjar, S. (Frank) Liu, *Adv. Sci.* **2022**, *9*, 2105103.
- [24] Q. Cao, Y. Li, Y. Zhang, J. Zhao, T. Wang, B. Yang, X. Pu, J. Yang, H. Chen, X. Chen, X. Li, S. Ghasemi, H. Salari, A. Hagfeldt, X. Li, *Adv. Energy Mater.* **2022**, *12*, 2201435.
- [25] H. Xu, Z. Liang, J. Ye, S. Xu, Z. Wang, L. Zhu, X. Chen, Z. Xiao, X. Pan, G. Liu, *Chem. Eng. J.* **2022**, *437*, 135181.
- [26] B. Yang, O. Dyck, J. Poplawsky, J. Keum, A. Puretzky, S. Das, I. Ivanov, C. Rouleau, G. Duscher, D. Geohegan, K. Xiao, *J. Am. Chem. Soc.* **2015**, *137*, 9210.
- [27] A. D. Jodowski, C. Roldán-Carmona, G. Grancini, M. Salado, M. Ralaifarisoa, S. Ahmad, N. Koch, L. Camacho, G. De Miguel, M. K. Nazeeruddin, *Nat. Energy* **2017**, *2*, 972.
- [28] X. Zhang, W. Zhou, X. Chen, Y. Chen, X. Li, M. Wang, Y. Zhou, H. Yan, Z. Zheng, Y. Zhang, *Adv. Energy Mater.* **2022**, *12*, 2201105.
- [29] G. Kim, H. Min, K. S. Lee, D. Y. Lee, S. M. Yoon, S. I. Seok, *Science* **2020**, *370*, 108.
- [30] M. Ledinsky, T. Schönfeldová, J. Holovský, E. Aydin, Z. Hájková, L. Landová, N. Neyková, A. Fejfar, S. De Wolf, *J. Phys. Chem. Lett.* **2019**, *10*, 1368.
- [31] X. Li, D. Bi, C. Yi, J.-D. Décoppet, J. Luo, S. M. Zakeeruddin, A. Hagfeldt, M. Grätzel, *Science* **2016**, *353*, 58.
- [32] N. K. Noel, A. Abate, S. D. Stranks, E. S. Parrott, V. M. Burlakov, A. Goriely, H. J. Snaith, *ACS Nano* **2014**, *8*, 9815.
- [33] E. Vega, M. Mollar, B. Marí, *J. Alloys Compd.* **2018**, *739*, 1059.
- [34] Y. C. Kim, K. H. Kim, D.-Y. Son, D.-N. Jeong, J.-Y. Seo, Y. S. Choi, I. T. Han, S. Y. Lee, N.-G. Park, *Nature* **2017**, *550*, 87.
- [35] H. Cho, S.-H. Jeong, M.-H. Park, Y.-H. Kim, C. Wolf, C.-L. Lee, J. H. Heo, A. Sadhanala, N. Myoung, S. Yoo, S. H. Im, R. H. Friend, T.-W. Lee, *Science* **2015**, *350*, 1222.
- [36] M. M. Tavakoli, M. Saliba, P. Yadav, P. Holzhey, A. Hagfeldt, S. M. Zakeeruddin, M. Grätzel, *Adv. Energy Mater.* **2019**, *9*, 1802646.
- [37] S. R. Cowan, A. Roy, A. J. Heeger, *Phys. Rev. B* **2010**, *82*, 245207.
- [38] L. Zhang, X. Zhang, X. Xu, J. Tang, J. Wu, Z. Lan, *Energy Technol.* **2017**, *5*, 1887.
- [39] S. Shao, Z. Chen, H.-H. Fang, G. H. Ten Brink, D. Bartsaghi, S. Adjokatse, L. J. A. Koster, B. J. Kooi, A. Facchetti, M. A. Loi, *J. Mater. Chem.* **2016**, *4*, 2419.
- [40] P. Wang, R. Li, B. Chen, F. Hou, J. Zhang, Y. Zhao, X. Zhang, *Adv. Mater.* **2020**, *32*, 1905766.
- [41] Q. Han, S. Bae, P. Sun, Y. Hsieh, Y. (Michael) Yang, Y. S. Rim, H. Zhao, Q. Chen, W. Shi, G. Li, Y. Yang, *Adv. Mater.* **2016**, *28*, 2253.
- [42] Y. Huang, L. Li, Z. Liu, H. Jiao, Y. He, X. Wang, R. Zhu, D. Wang, J. Sun, Q. Chen, H. Zhou, *J. Mater. Chem.* **2017**, *5*, 8537.
- [43] F.-Z. Qiu, M.-H. Li, S. Wang, Y. Jiang, J.-J. Qi, J.-S. Hu, *J. Mater. Chem. A* **2022**, *10*, 3513.
- [44] R. H. Bube, *J. Appl. Phys.* **1962**, *33*, 1733.
- [45] B. Liu, H. Bi, D. He, L. Bai, W. Wang, H. Yuan, Q. Song, P. Su, Z. Zang, T. Zhou, J. Chen, *ACS Energy Lett.* **2021**, *6*, 2526.
- [46] D. Yang, R. Yang, K. Wang, C. Wu, X. Zhu, J. Feng, X. Ren, G. Fang, S. Priya, S. Liu, *Nat. Commun.* **2018**, *9*, 3239.
- [47] M. Othman, T. Zhang, D. P. McMeekin, S. O. Furer, W. Mao, W. Li, A. D. Scully, A. S. R. Chesman, P. N. H. Nakashima, U. Bach, J. Etheridge, *Sol. RRL* **2023**, *7*, 2200852.